



GE Medical Systems

Technical Publications

Direction 2257633
Revision 0

Signa® Release 8.3 Software Upgrade - Version M4

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Operating Documentation

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3/12/92



GE Medical Systems

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- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
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- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
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ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

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- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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- 本维修手册仅存有英文本。
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1- OVERVIEW

This manual covers upgrade of a Signa Release 8.2.5 system to Release 8.3. The Release 8.3 upgrade includes software and hardware. This manual is delivered with two catalogs:

- M1033KA – Release 8.3 upgrade for systems with Indigo Computers
- M1033KB – Release 8.3 upgrade for systems with Octane Computers

1-1 Furnished Materials

<u>ITEM</u>	<u>PART NUMBER</u>	<u>DESCRIPTION</u>	<u>QTY.</u>
1	2226715	2 nd Hard Drive (for Indigo only)	1
2	2149503	Memory, 72 SIM (for Indigo only)	4
3	2199806-5	Memory, 2 x 128 Mbyte (for Octane only)	1
4	2254693	LX Tools Key	1
5	2156158-16	Release 8.3 Software Collector	1
5a	2204244-10	Application CD-ROM	1
5b	2256092	IRIX 6.5.4 CD-ROM set (2 disk OS; 1 Boot/Update)	3 disk set
5c	2214164	MaxOptix MOD Media blank	1
6	46-302200P3	Modification Rating Plate	1
7	46-170015P10	Screw (for Indigo Memory boards)	4
8	2209595	Interactive Vascular Software MOD	1
9	2209382	Horizon LX Software Options Install	1
10	2249665	Castenet Software License package	1
11	2240540	Netscape License package	1
12	2257633	<i>This document</i>	1

1-2 Manpower Requirements

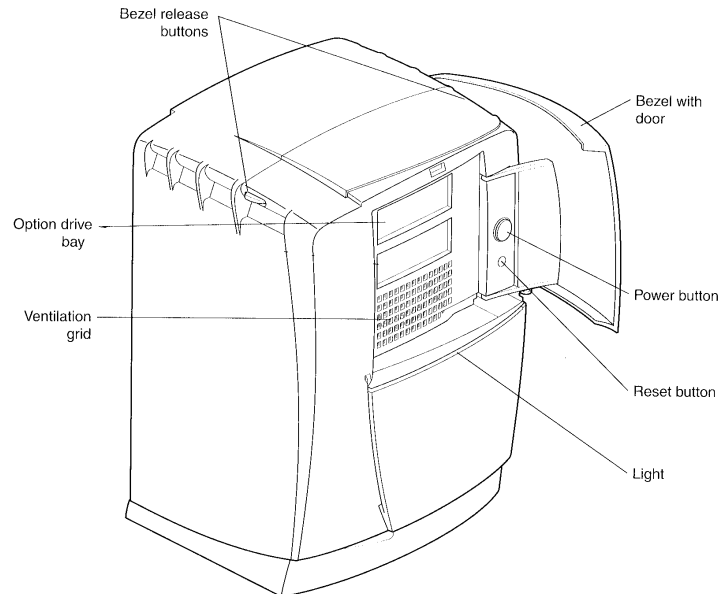
One Service Engineer for three (3) hours for software install (no calibrations required). Plus one-quarter hour for SGI computer memory upgrade. Plus (for SGI Indigo sites only) one additional hour required for installing a second hard drive.

1-3 Special Tools & Test Equipment

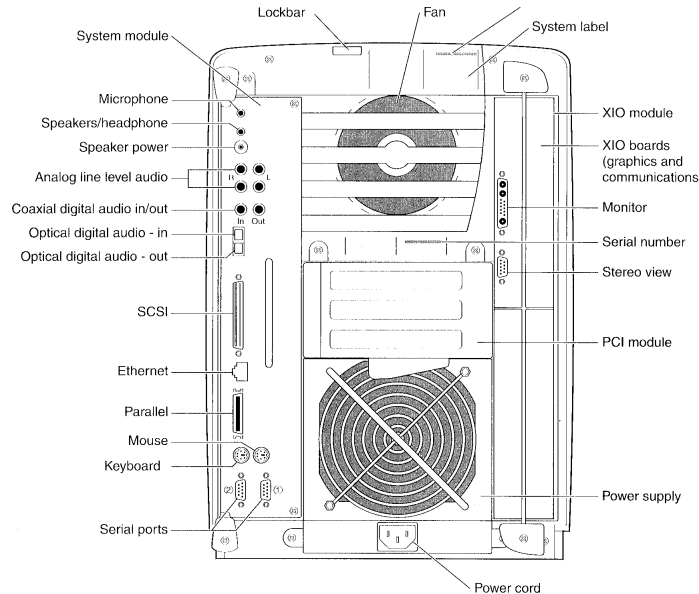
None

2- MEMORY UPGRADE – SGI OCTANE COMPUTERS ONLY

This section provides instructions to install the additional 256 Mbyte of memory to the Octane computer. The Octane computer is now located on the workspace table. For front and back view illustrations of the SGI OCTANE Tower, see Illustrations 2-1 and 2-2.



SGI TOWER (FRONT VIEW)
ILLUSTRATION 2-1



SGI TOWER (REAR VIEW)
ILLUSTRATION 2-2

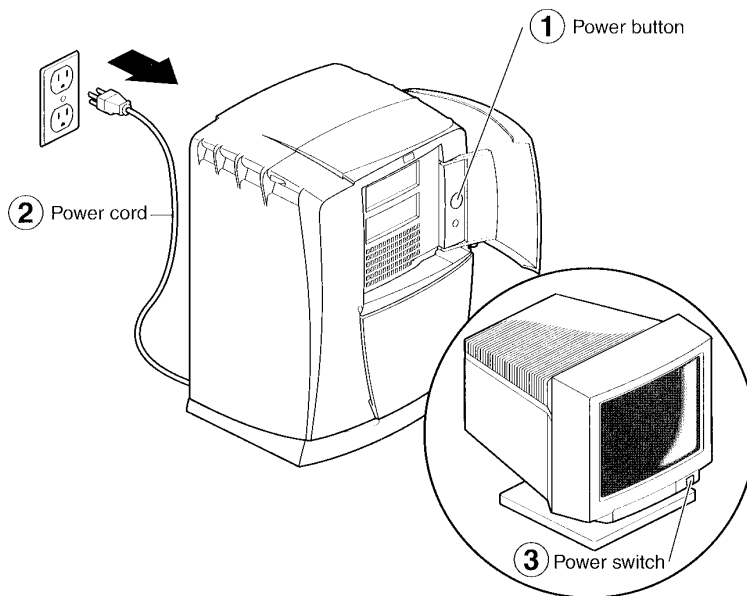
2-1 Preparing the System for Installation or Removal of Replacement Parts

Several steps need to be taken prior to actual removal or installation of replacement parts. This will ensure operator safety.

2-2 Powering Off the OCTANE Workstation

You must first follow proper procedures to shut down any programs you are working in and log out from the Host Computer.

1. Log out from Host Computer.
2. Shut down the Host Computer.
3. At the PDU, open the circuit breaker(s) that power the Operator Workspace to turn power off.
4. Lock and tag the circuit breaker(s). (Refer to CD-ROM *Dir. 2160623-1 [or -2], Signa Release 8x Service Methods*, Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
5. Verify that power is off by twice pressing (on/off) the Host Computer power button; the LED should remain off.
6. Open the front cover of the workstation and verify power is off by twice pressing (on/off) the power button to the OCTANE workstation. See Illustration 2-3.



OCTANE POWER BUTTON LOCATIONS
ILLUSTRATION 2-3

7. Unplug the power cord from the electrical outlet and from the OCTANE workstation.

WARNING!

SOME POWER STILL REMAINS AFTER POWER OFF USING THE POWER SWITCH. TO ENSURE THAT ALL POWER IS REMOVED, UNPLUG THE COMPUTER.

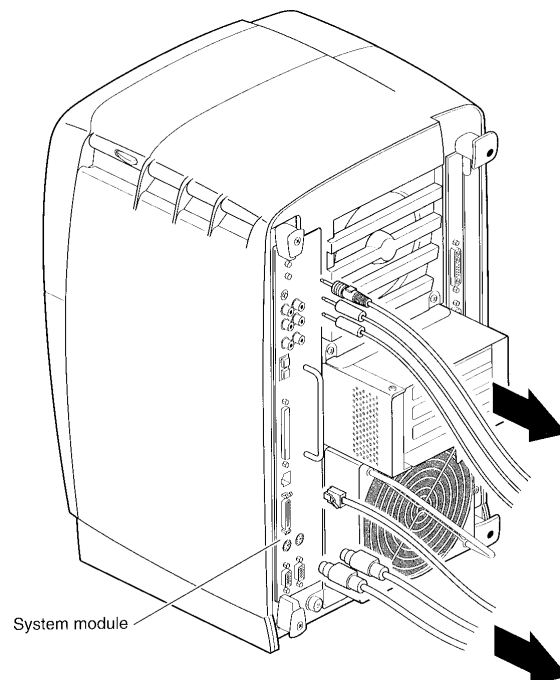
8. Turn off monitor by pressing the monitor power button. See Illustration 2-3.
9. Wait five minutes for workstation to cool down.

WARNING!

THE HEAT SINKS ON THE CPU (PROCESSOR) AND SYSTEM MODULE BECOME VERY HOT. WAIT FIVE MINUTES AFTER POWERING OFF THE OCTANE WORKSTATION BEFORE YOU REMOVE THE ANY COMPONENTS. TEST BEFORE TOUCHING THE CPU OR HEAT SINKS.

2-3 Removing the System Module

1. Remove cables attached to the System Module. See Illustration 2-4.



SYSTEM MODULE CABLES
ILLUSTRATION 2-4

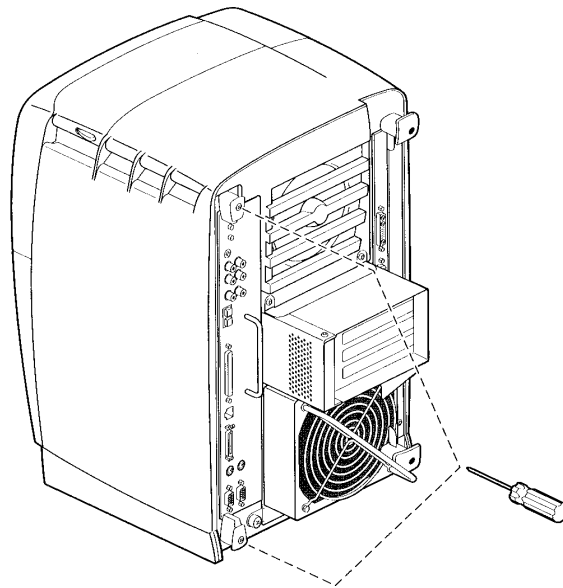
CAUTION

The components inside the OCTANE workstation are extremely sensitive to static electricity; you must wear a wrist strap while replacing parts inside the workstation.

WARNING!

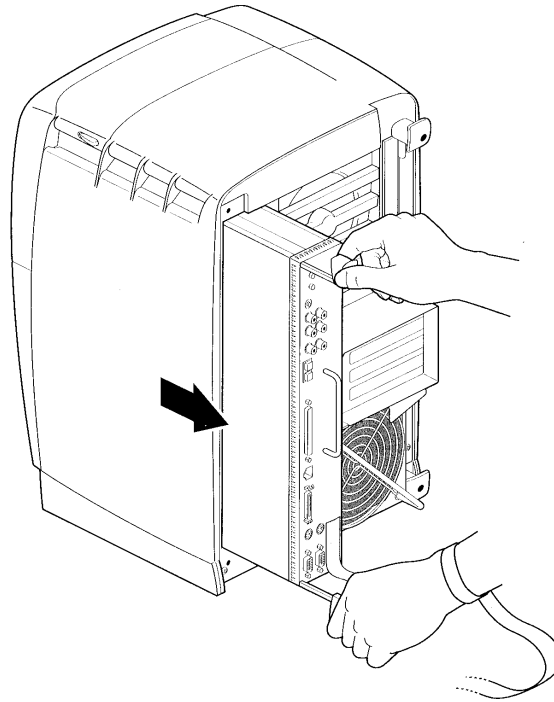
THE HEAT SINKS ON THE CPU (PROCESSOR) AND SYSTEM MODULE BECOME VERY HOT. WAIT FIVE MINUTES AFTER POWERING OFF THE OCTANE WORKSTATION BEFORE YOU REMOVE THE SYSTEM MODULE. TEST BEFORE TOUCHING THE CPU OR HEAT SINKS.

2. To ensure operator safety, be sure the power is off, and that five minutes have elapsed to allow the heat sinks to cool, and you have attached a wrist strap. The System Module must be removed to install memory.
3. Locate and loosen the System Module screws that hold the sliding handles to the OCTANE workstation. See Illustration 2-5.



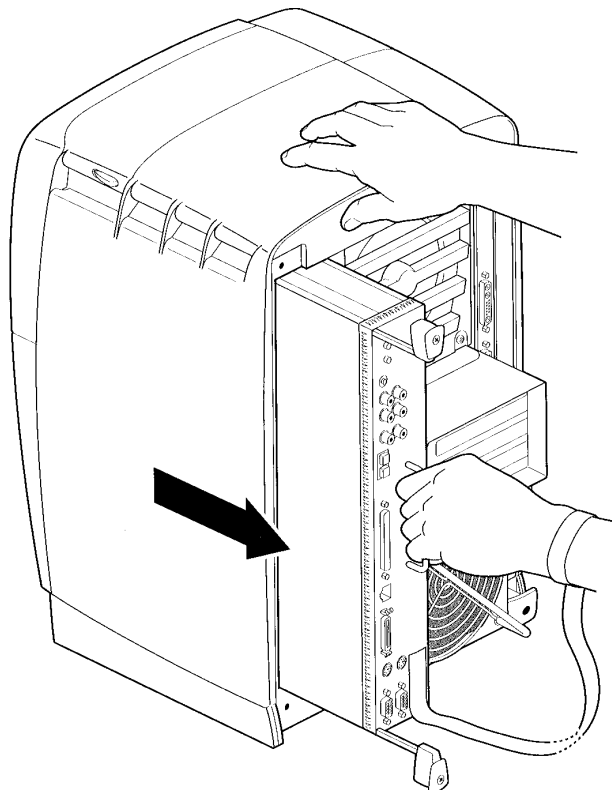
SYSTEM MODULE SCREWS
ILLUSTRATION 2-5

4. Pull both sliding handles simultaneously until they are completely extended to release the System Module from the workstation. See Illustration 2-6.



SLIDING OUT SYSTEM MODULE
ILLUSTRATION 2-6

5. Pull the System Module from the chassis by grasping module handle with one hand and bracing your other hand against the top of the OCTANE workstation. See Illustration 2-7.



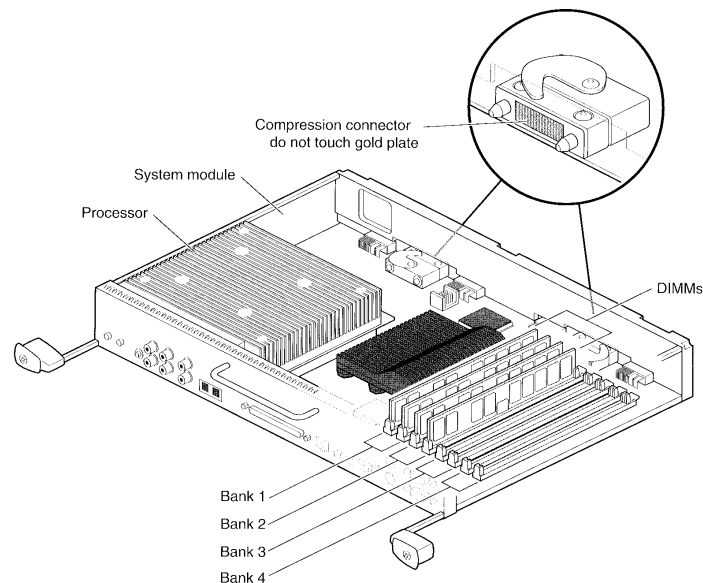
REMOVING SYSTEM MODULE FROM CHASSIS
ILLUSTRATION 2-7

6. Place System Module on a dry, antistatic surface such as your desktop. Keep the CPU and Dual inline memory modules (DIMMs) facing up.



Dual inline memory modules, or DIMMs, are extremely sensitive to static electricity. Handle the DIMMs carefully and wear a wrist strap to avoid the flow of static electricity.

7. Locate the DIMM banks and sockets. See Illustration 2-7.

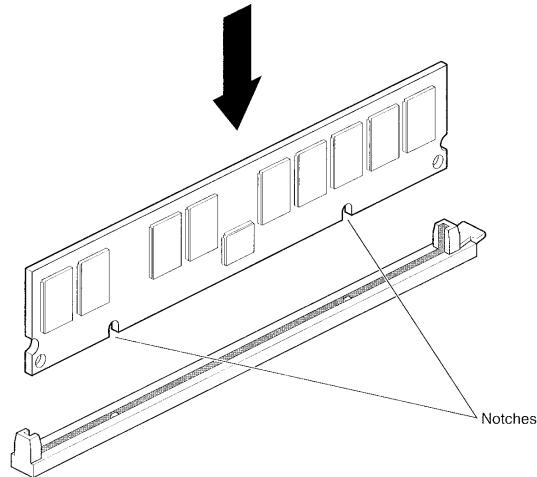


DIMM BANKS AND SOCKET LOCATIONS
ILLUSTRATION 2-7



Do not touch the connector near the DIMM removal levers.

8. DIMMs are notched on the bottom so that they cannot be inserted incorrectly. See Illustration 2-8.
9. Insert the pair of DIMM boards into Bank 3. Listen for a click noise as it each DIMM board makes its connection and the latch on the end of the socket moves up.

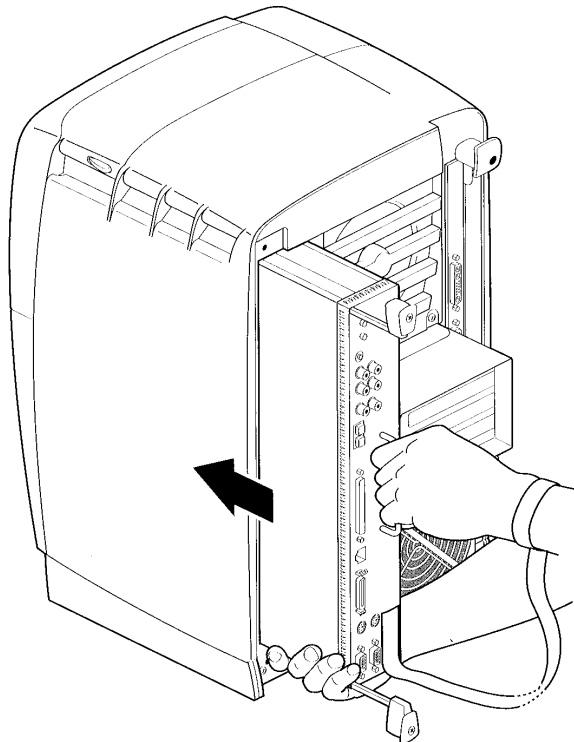


INSTALLING DIMMS
ILLUSTRATION 2-8

10. The memory installation is complete and the System Module can now be reinstalled.

2-4 Reinstalling the System Module

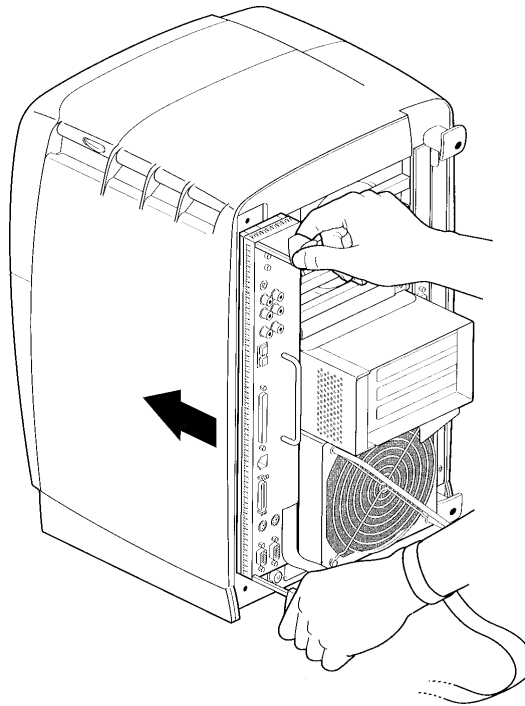
1. Grasp the System Module by its handle and support it with one hand as you slide it into the chassis. See Illustration 2-9.



REPLACING THE SYSTEM MODULE
ILLUSTRATION 2-9

2. Push the module completely into the chassis using the handle. Both sliding handles protrude from the ends of the System Module. The System Module stops when it is slightly out of the chassis.

3. Push the sliding handles simultaneously to connect the System Module to the workstation.
See Illustration 2-10.



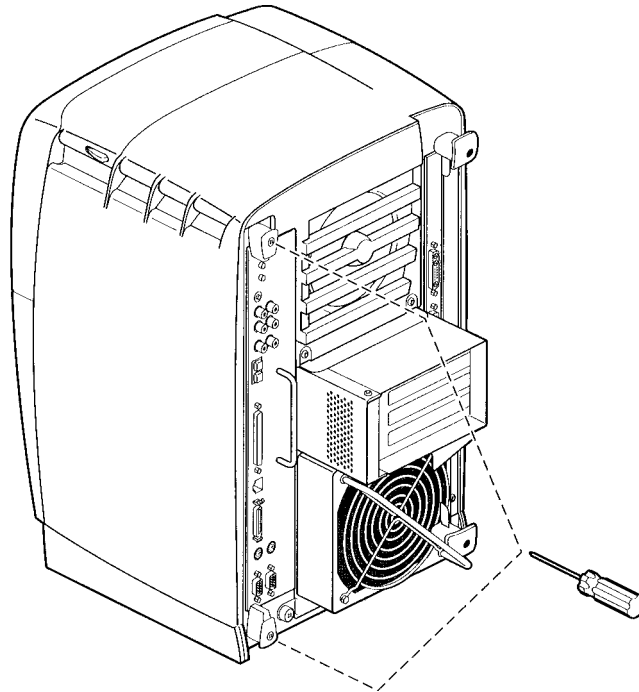
SLIDING IN SYSTEM MODULE
ILLUSTRATION 2-10

4. Push firmly on the sliding handles to completely lock the system module to the workstation.

Note

The sliding handles are completely recessed and the system module is flush with the chassis when it is fully positioned and locked to the workstation.

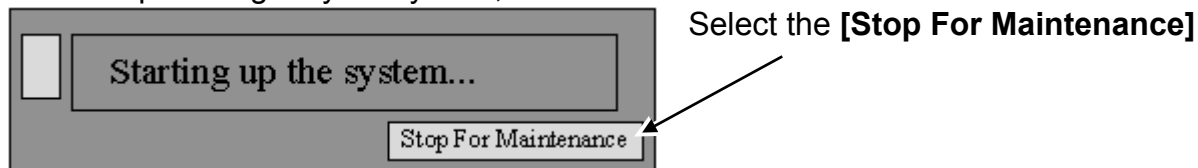
5. Tighten the connected screws in the sliding handles until the system module is attached to the chassis. See Illustration 2-11.



REPLACING SYSTEM MODULE SCREWS
ILLUSTRATION 2-11

2-5 Verifying Memory Installation

1. After powering on your system,



2. Select **[Stop For Maintenance]**. Click on **[Command Monitor]** when it becomes available.

Type the following: **hinv <Enter>**

The System Manager should report 512 Mb (256 Mb original + 256 Mb just added). If the reported amount of memory is incorrect, power off the system and confirm the installation or removal process.

- Check to see that the angle of the DIMMs are upright and completely inserted.
- Check that each bank is occupied with two DIMMs and they are the same type. You must have an even amount of DIMMs installed.
- Check the label color on the DIMM. Both DIMMs in a bank must have the same colored label.

3. Proceed to Section 4- Prep for Software Installation – Both Octane and Indigo Computers.

3- HARDWARE INSTALLATION – SGI INDIGO COMPUTERS ONLY

This section provides instructions to install a second (4 Gbyte) Hard Drive and additional 128 Mbyte of memory to the Indigo computer.

3-1 Second Hard Drive Installation – Required for SGI Indigo Computers Only

This section describes the installation of the 4.0 Gbyte hard drive (part # 2226715).

Tools Note: you may need extra jumpers for disk drive, phillips screwdriver, needle nose pliers.

1. Select [**Service Desktop Manager**], then [**CShell**].
2. With the cursor in the C Shell Window type:
hinv <Enter>
Iris Audio Processor: version A2 revision 1.1.0
1 250 MHZ IP22 Processor
FPU: MIPS R4000 Floating Point Coprocessor Revision: 0.0
CPU: MIPS R4400 Processor Chip Revision: 6.0
On-board serial ports: 2
On-board bi-directional parallel port
Data cache size: 16 Kbytes
Instruction cache size: 16 Kbytes
Secondary unified instruction/data cache size: 2 Mbytes on
Processor 0
Main memory size: 256 Mbytes
EISA bus: adapter 0
Integral Ethernet: et0, I00
Integral Ethernet: ec0, version 1
Integral SCSI controller 1: Version WD33C93B, revision D
Integral SCSI controller 0: Version WD33C93B, revision D
Disk drive / removable media: unit _____ on SCSI controller 0
Optical disk: unit _____ on SCSI controller 0
CDROM: unit _____ on SCSI controller 0
Disk drive: unit _____ on SCSI controller 0
Graphics board: High Impact/TRAM option card
3. Fill in the blanks (above) for current unit numbers.
4. Type: **exit <Enter>**
5. From the Service Desktop, select [**System Shutdown**], then [**Yes**].
6. Logoff the PC.
7. Power down the Indigo II Host Computer.
8. As applicable, open the circuit breaker(s) that power the Operator Workspace (OW) to turn power off.
9. Lock and tag the circuit breaker(s).

Note

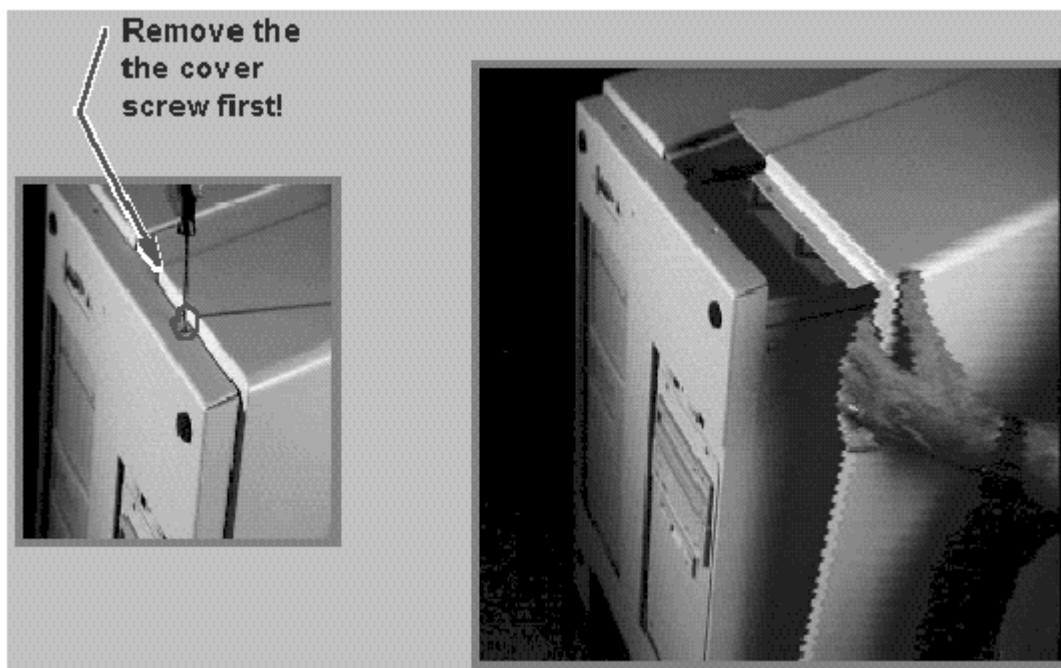
Use proper ESD precautions when working on this hardware. These components are ESD sensitive and damage will occur to the hardware if proper ESD measures are not followed. Always use appropriate ESD procedures and an ESD Kit. Use a grounding wrist strap and ESD Storage Bag.

10. Grasp the front cover from the sides (preferred) or from the top and bottom.
11. Pull the cover straight toward you. The ball and socket clips should release with as little as 8 lbs. (2.984 kg) of pressure. See Illustration 3–1.



OW CABINET FRONT COVER REMOVAL
ILLUSTRATION 3–1

12. Loosen the rear thumbscrew but do not remove it.
13. Remove the Phillips head screw at the top of the front cover.
14. Lift the bottom of the cover slightly and slide the cover to the rear of the cabinet about 1/2" (13mm).
15. Pull the cover toward you until the tabs clear the two top slots. See Illustration 3–2.



OW CABINET: SGI HOST SIDE COVER REMOVAL
ILLUSTRATION 3-2

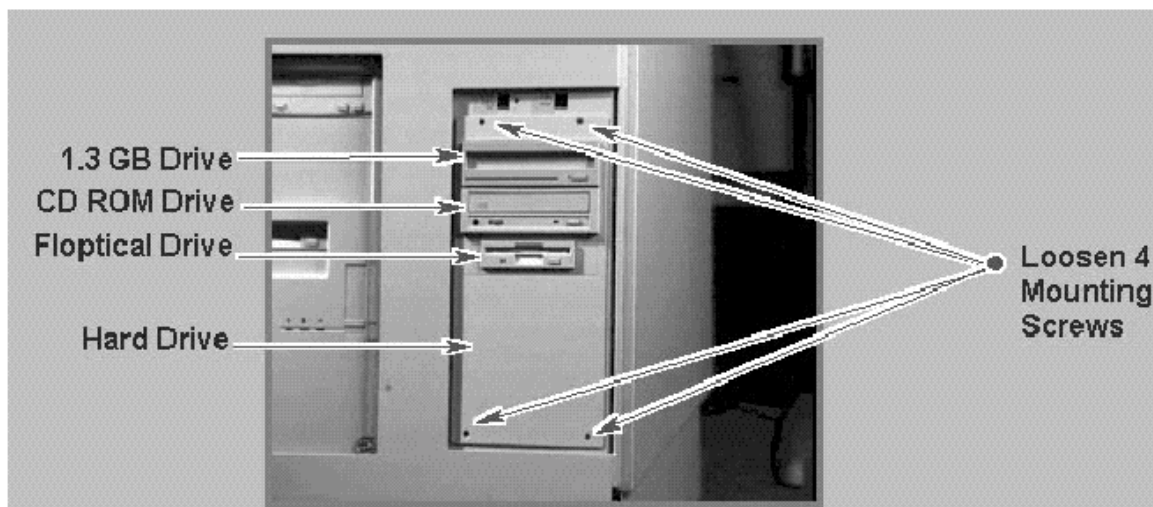
Note

Prior to removal of any internal or external cables including power cords, label all cables and corresponding ports to assist in reinstallation.

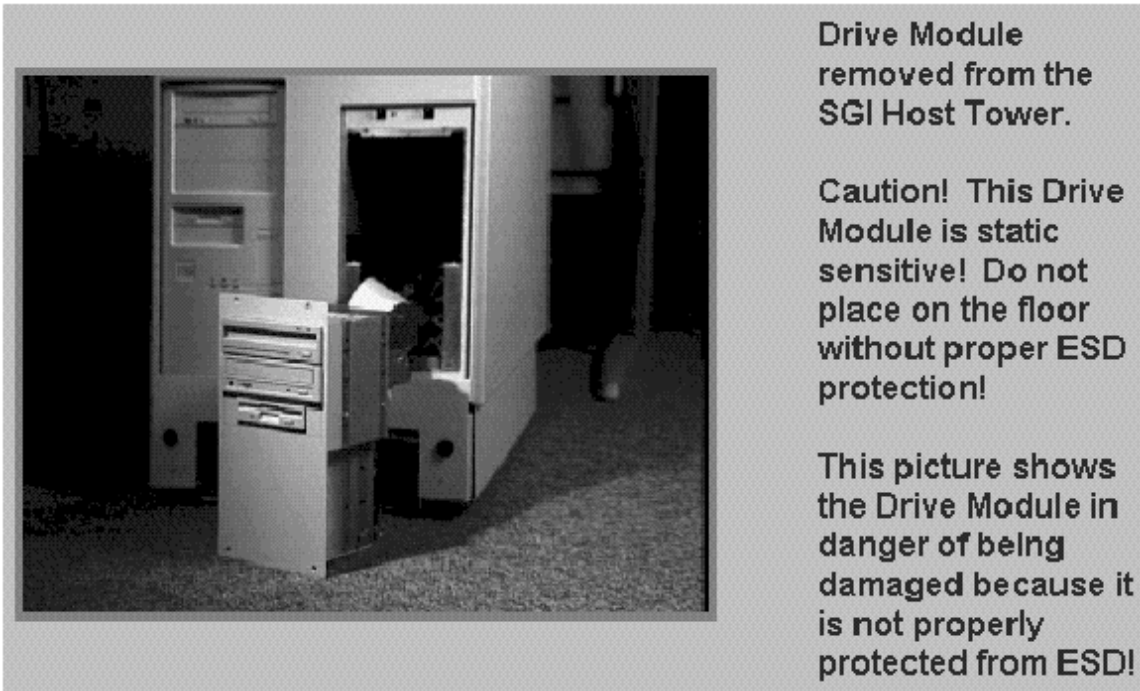
CAUTION

Use caution when removing panels and modules from the Operator Workspace. ESD copper fingers located on the inside edges of the SGI housing and cover panels damage easily. Handle carefully to prevent damage to your hands and the ESD fingers. If damage occurs to the ESD fingers, they must be replaced in their slot like zipping a zipper. Start where the fingers are aligned correctly and press the damaged fingers in place by sliding your fingers along the edges as applicable.

16. On the front of the SGI Host Tower, unscrew the four mounting screws. See Illustration 3-3.
17. Gently pull the drive housing out and away from the SGI Host Tower. Unless you are servicing the individual drives, do not disconnect the drive housing cables from their internal connections. Cables are long enough to extend outside the Tower. See Illustration 3-4.

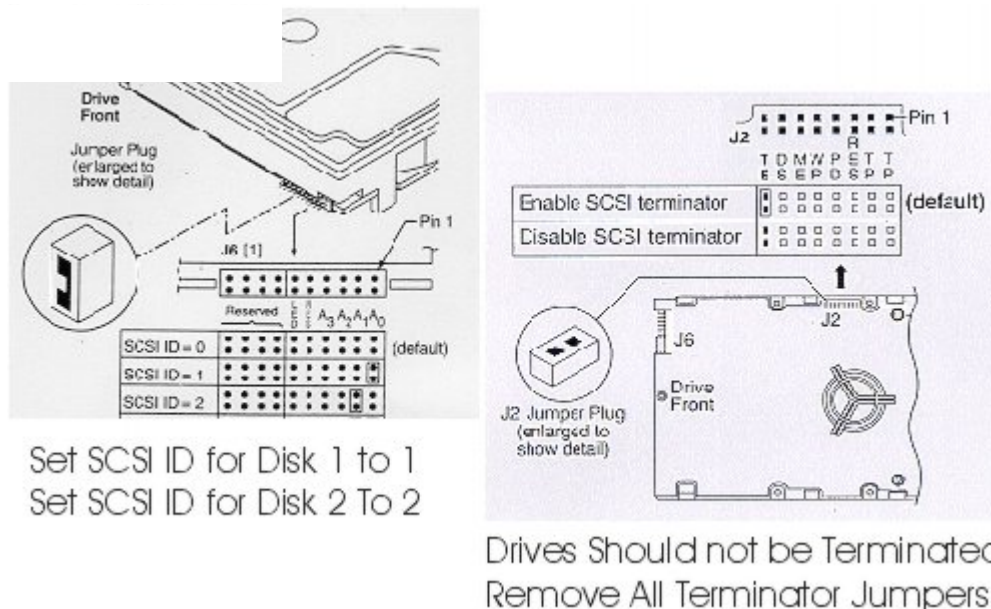


SGI HOST TOWER: DRIVE MODULE REMOVAL –1
ILLUSTRATION 3-3



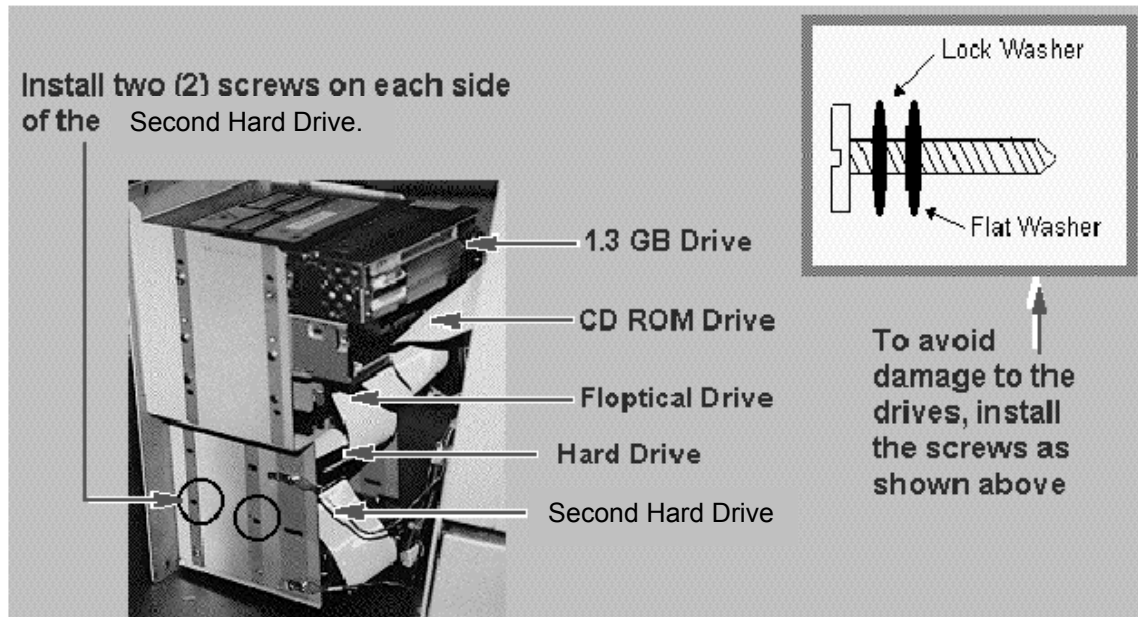
SGI HOST TOWER: DRIVE MODULE REMOVAL – 2
ILLUSTRATION 3-4

18. Remove the 2nd Hard Drive from the ESD storage bag. Label this as unit 2.
19. **Configure** the 2nd Hard Drive as **SCSI device = 2**. See Illustration 3-5.



SEAGATE DISK DRIVE JUMPER SETTING
ILLUSTRATION 3-5

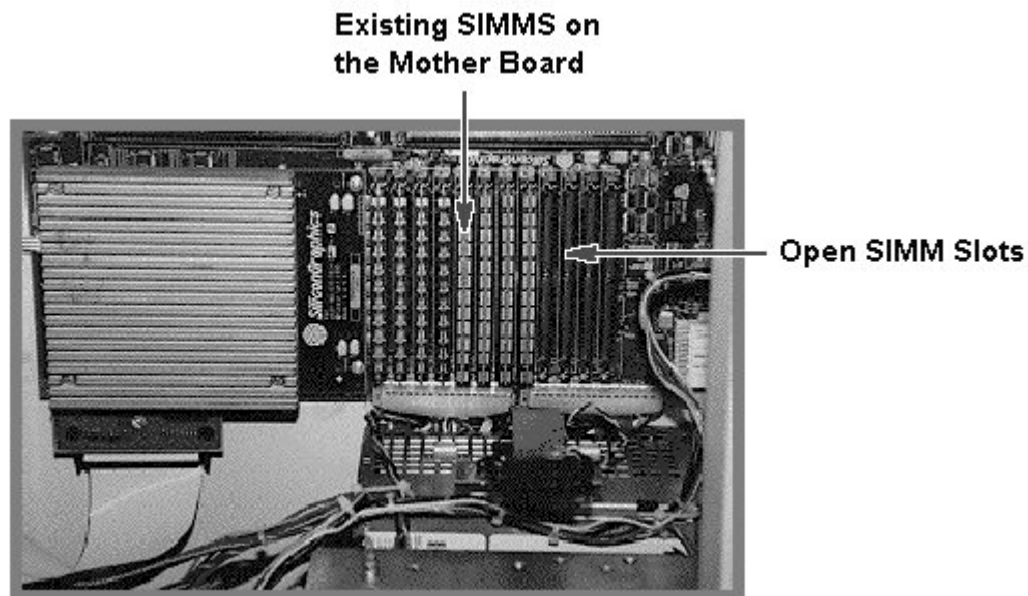
20. **If present re-configure** the Floppy Drive as **SCSI device = 5**.
21. Slide the Hard Drive into the 2nd from the bottom slot at the rear of the Drive Housing. Raise the Hard Drive slightly to align the screw holes with the bracket. See Illustration 3–6.
22. Screw in the two slotted screws on either side of Hard Drive (four total) to attach the drive to the housing. See Illustration 3–6.
23. At the rear of the Drive Housing, connect one of the extra SCSI cable connectors to the 2nd Hard Drive.
24. At the rear of the Drive Housing, connect one of the extra power connector cables to the 2nd Hard Drive. The cable has a small white connector attached to red, black and orange wires.



SCSI CABLE DETAIL ON THE DRIVE MODULE
ILLUSTRATION 3-6

3-2 Memory Upgrade - Required for SGI Indigo Computers Only

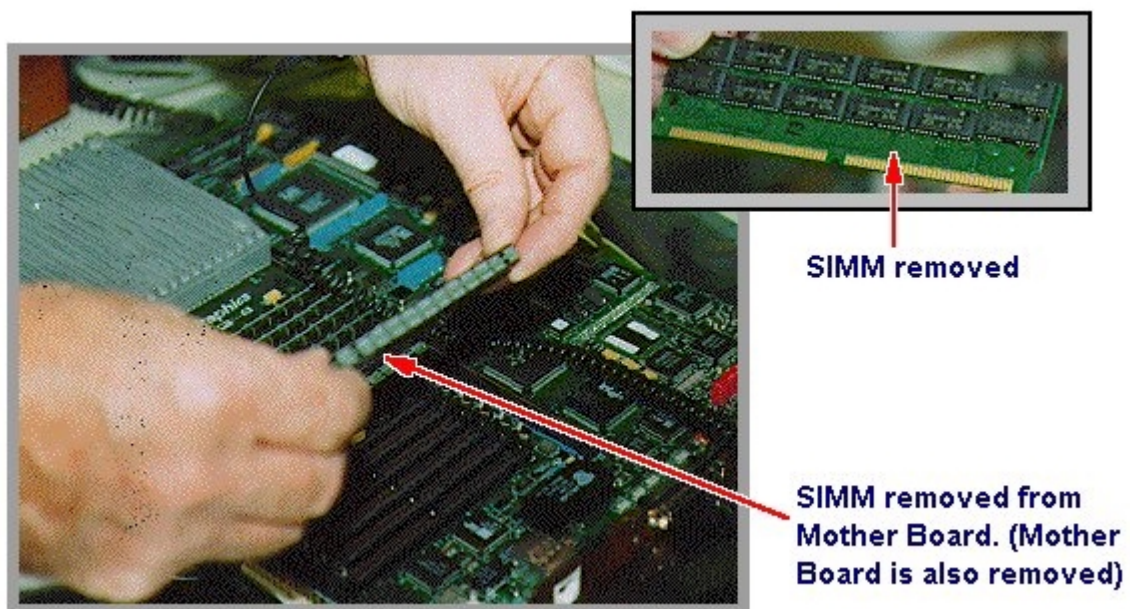
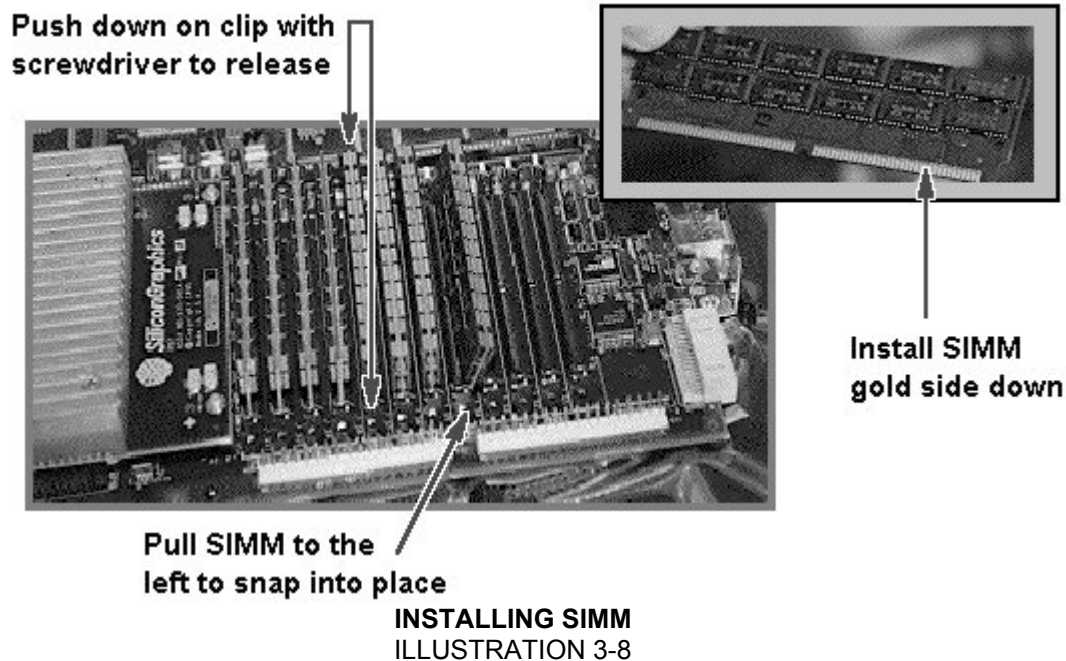
From the side of the SGI Tower, locate the SIMM slots on the right side of the Mother Board. See Illustration 3-7.



SIMM LOCATION
ILLUSTRATION 3-7

3-2-1 Memory Installation Procedures

1. Remove the additional SIMMs from the original carton, as applicable.
2. Locate the SIMM slots on the right side of the Mother Board. See Illustration 3-8.
3. Gently push the SIMM (gold side down) into an open slot. The SIMM installs on slight angle. See Illustration 3-9.
4. Pull the SIMM to the left until it snaps into place.



3-2-2 Restore Indigo

1. Reverse the removal procedures.
2. Ensure the PC side cover is replaced first.
3. Make sure all screws are installed before replacing the OW Cabinet – SGI (Host) side – cover.
4. Remove Lock and tagged circuit breaker(s).
5. As applicable, close the circuit breaker(s) that power the Operator Workspace.
6. Boot up the system.

3-3 Second Hard Drive Verification

1. Select [**Service Desktop Manager**], [**CShell**].
2. With the cursor in the C Shell Window type:
hinv <Enter>
Iris Audio Processor: version A2 revision 1.1.0
1 250 MHZ IP22 Processor
FPU: MIPS R4000 Floating Point Coprocessor Revision: 0.0
CPU: MIPS R4400 Processor Chip Revision: 6.0
On-board serial ports: 2
On-board bi-directional parallel port
Data cache size: 16 Kbytes
Instruction cache size: 16 Kbytes
Secondary unified instruction/data cache size: 2 Mbytes on
Processor 0
Main memory size: 384 Mbytes **<– (Verify 384 Mbytes total now shown)**
EISA bus: adapter 0
Integral Ethernet: et0, I00
Integral Ethernet: ec0, version 1
Integral SCSI controller 1: Version WD33C93B, revision D
Integral SCSI controller 0: Version WD33C93B, revision D
Disk drive / removable media: unit 5 on SCSI controller 0
Optical disk: unit 4 on SCSI controller 0
CDROM: unit 3 on SCSI controller 0
Disk drive: unit 2 on SCSI controller 0 **<– 2nd Hard Drive**
Disk drive: unit 1 on SCSI controller 0
Graphics board: High Impact/TRAM option card
3. Verify the 2nd Hard Drive is **unit 2 on SCSI controller 0** and Main memory size is 384 Mbytes.
4. Type: **exit <Enter>**.

4- PREP FOR SOFTWARE INSTALLATION – BOTH OCTANE AND INDIGO COMPUTERS

4-1 Archive Images

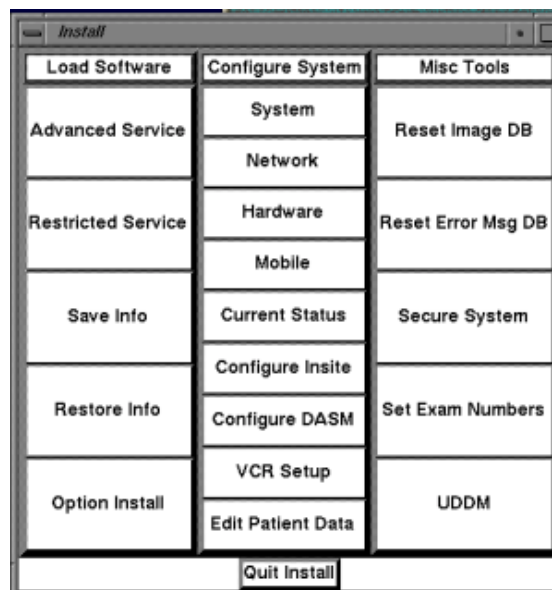
This installation instruction will erase any current images left in the database. **Archive all images BEFORE continuing with this procedure.**

4-2 Site Information

For systems with an SGI Indigo computer that have not installed a second disk drive, refer to Section 3, Hardware Installation – SGI Indigo Computers Only to physically install a second disk drive that was shipped with this upgrade package. Following installation of the second drive, proceed below to load the host software.

4-2-1 Saving System Configuration Files, Proprietary Tools/Diags, and Site/GE Protocols

1. Click on the **[Utilities]** icon from the Service Desktop control panel.
2. From the Service Desktop select **[Install]**. The Install Tool window opens.
3. In the Install Tool window, type **operator <Enter>** at the password prompt. The Install menu will open (see Illustration 4-1).



INSTALL MENU
ILLUSTRATION 4-1

4. Put the INFO MOD into the MOD with side A up.
5. Click on **[Save Info]** from the Install menu. The system opens the window called Xexec.

Note

If the MOD has not been used, you will see a message regarding the use of the MOD for “Raw Data Storage”; choose **[Overwrite]**.

6. When asked Do you really want to save configuration files to the INFO MOD? [y|n], type: **y <Enter>**
7. SaveInfo is now complete. Remove the MOD media and write the date and time on it.

4-2-2 If Save/Restore Functions Fail Because The Host Can Not See The MOD Drive.

If the system MOD, Floppy or CD-ROM drives are changed in the system, configuration maps may no longer know that these devices exist. If Save / Restore functions do not work because the system say's no MOD device exists, do the following:

1. Click on the **[System Shutdown]** button on the Service Desktop control panel.
2. Restart the system after it shuts down and log in as root, password operator.
3. From the Root menu bar on the right hand side of the desktop, select **[Reconfigure]** by clicking the right mouse button over the reconfigure option.
4. Left mouse button click the option **[Configure Devices]**. A “Term” window will now open and remain blank for up to two minutes. When finished, “Done with Device Configuration” will appear in the Term window. Press any key to exit. Reboot the system to **[Signa]** and continue with this section.
5. Reboot the system before continuing with the following steps. Failure to reboot may cause the following steps to not work and system information will be lost.

Other SaveInfo problems

Problem 1: SaveInfo may incorrectly identify the Info MOD disk as non-info MOD disk.

Solution 1: Be extra careful and make sure that the disk does not contain any important data, which can be overwritten when the option is chosen.

Problem 2: While saving information to the disk, “tar:tape write error: No space left on device” or some other error may appear.

Solution 2: Ignore this error message; SaveInfo will continue saving the files.

4-2-3 Required Information Prior To Loading Software

First collect vital site information listed below to insure the install will not be stopped for lack of site specific information. To collect this info: On the Service Desktop, select **[Install]**, type **operator** <Enter> for the password. Look in Hardware, Network and System Configuration buttons for current site information. Refer to Appendix B for instructions on how to locate the site information. Use Table 4-1 below to record your Site configuration information.

- Unique Internet IP Address for Host (Networked systems only).
- Unique Internet IP address for PC Computer.
- Unique Internet IP Address for Laptop port.
- Netmask
- Domain Name *
- Modem Type *
- Hostname (Unique name for this system. Name must begin with Alpha character. Beginning the name with a number will cause the system to not come up.).
- System Type (Echospeed, Highspeed, Horizon) *
- Unique ID (from GECARES specific to site)
- Service ID (from GECARES, assigned when first dispatch is opened on system)
- Suite Name: This name should be at least, but no more than 4 characters.
- Camera name *
- DASM type *
- Field Strength
- Gradient Type

* These entries are not found in the Install GUI. They are acquired by physically checking the system configuration.

TABLE 4-1
VITAL SITE INFORMATION

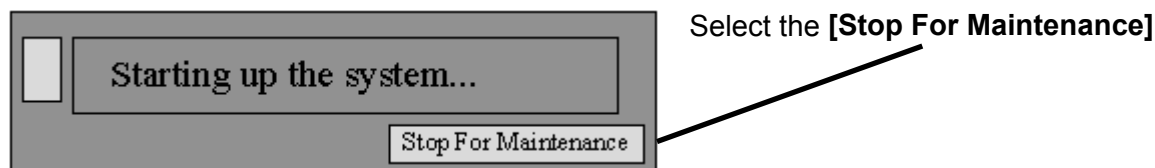
ENTRY	VALUE	ENTRY	VALUE
Host IP:		Host ID:	
Netmask:	Default is 255.255.252.0	Unique ID:	
Domain Name		Coil Type	☐ BRM ☐ CRM
IWS PC IP:		RF Type	☐ Tube Type RF PEN II ☐ 1.0T Teli Solid State RF ☐ 1.5T Analogic Solid State RF
FE Laptop IP:		Service ID:	
TPS Subnet IP:		Weight Unit:	
Network Speed		Field Strength:	☐ 1.0T ☐ 1.5T
Host Name:	Note: Name can not begin with a number.	Gradient Type:	☐ 8645 Single or Double Bay ☐ 8280 SGD Base System ☐ 8651 SGD HighSlew
Hopsital Name:		Camera Type:	
Suite ID:		DASM Type:	☐ Digital (LCAM) ☐ Video (VDB)
Package Name:	☐ Echosped SR150 ☐ Echosped SR120 ☐ Highspeed SR77 ☐ Base (Horizon) SR20	Receiver BW	☐ 62KHz ☐ 125KHz

A SaveInfo should be done before starting the install is started to insure the latest copy of current system settings. If this is software upgrade on a system already at 8.3_M3 or greater, create a config MOD from the Guided Install File menu. Insert the your SaveInfo MOD or new MOD into the drive. Go to the top left corner of the Install GUI and select **<File>** and then **<Save to MOD>**. This process does not create a SaveInfo disk. It just creates a copy of the information already entered in the GUI tabs for later use for updating the tabs in a new install. SaveInfo is done by going to the Save/Restore tab. The **<Save to MOD>** and SaveInfo can live on the same MOD. A separate partition is built for each.

It is critically important to insure the correct values are entered for System type, Gradient Type, Field Strength, RF Type and Coil Type before doing the MRApps load. These values determine what protocols and coil configurations files are loaded.

5- HOST OPERATING SYSTEM SOFTWARE LOAD

1. With the system up and running, go to the Service Desktop and select **[System Shutdown]**. If the system is not up and running, switch on power at this time. When prompted, click on the **[Restart Button]**.
2. Stop the reboot process by clicking on the **[Stop For Maintenance]** button when it appears.



3. Click on the **[Enter Command Monitor]** button when it becomes available.
4. Insert the **IRIX 6.5** CD labeled **“Boot/Update”** into the CDROM Drive on the workstation (right side). On the command line type the following:

Note

User responses are bolded text.

Depending on the host computer type your system is using, boot the CD by typing one of the following:

If you are loading to an SGI Octane computer:

```
>> boot -f dksc(1,3,8)sash64 dksc(1,3,7)stand/fx.64 - -x <Enter>
```

If you are loading to an SGI Indigo computer:

```
>> boot -f dksc(0,3,8)sashARCS dksc(0,3,7)stand/fx.ARCS - -x <Enter>
```

Note

The rest of the install is for all SGI computer types.

```
fx: "device-name" = (dksc) <Enter>
```

```
.  
fx: ctrl# = (0) <Enter>
```

```
.  
fx: drive# = (1) <Enter>
```

```
.  
fx> r/ro <Enter>
```

```
.  
fx/Repartition/RootDrive: Type of Data Partition = (xfs) <Enter>
```

```
.  
fx> continue? yes <Enter>
```

```
.  
fx> l/sy <Enter> (This is a lower case L and / lower case s and lower case y)  
.  
fx> exit <Enter>
```

At this point the system will reboot to the PROM startup screen.
Select **[Install System Software]** button. A pop up window opens asking where to install from.
Choose **[Local CD-ROM]**; select **[Install]**; then select **[Continue]**.

If you are working with newly formatted disk drives or new, never used disk drives, you will be told by the system during this boot that the system was unable to mount the "/root" file system and the system will stop with the statement "press Enter to invoke C shell csh:".

If you see this message, type the following:

...press Enter to invoke C shell csh: **<Enter>**

mkfs /dev/dsk/dks0d1s0 (This is /dev/dsk/dks zero d one s zero)

exit <Enter>

5. The next system message will be:

If you are working with a newly formatted drive you may get a message like:
*/dev/dsk/dks0d1s0: Invalid argument No valid file system found on:
/dev/dsk/dks0d1s0 This is your system disk: without it we have nothing on which
to install software.*

Do the following.

Make new file system on /dev/dsk/dks0d1s0 [yes/no/sh/help]: **yes<Enter>**

About to remake (mkfs) file system on: /dev/dsk/dks0d1s0.

This will destroy all data on disk partition: /dev/dsk/dks0d1s0.

Are you sure? [y/n] (n): **y<Enter>**

Block size of filesystem 512 or 4096 bytes? **512<Enter>**

Doing: mkfs -b size=4096 /dev/dsk/dks0d1s0

meta-data=/dev/rdisk/dks0d1s0 isize=256 agcount=8, agsize=128612 blks

data = bsize=4096 blocks=1028895, imaxpct=25

log =internal log bsize=4096 blocks=1000

realtime =none bsize=65536 blocks=0, rtextents=0

Trying again to mount /dev/dsk/dks0d1s0 on /root.

inst> **sh** <Enter>

.

mount /CDROM <Enter>

.

/CDROM/mr_prep <Enter>

.

6. The system will run for about 6 minutes and then will prompt with:

"Please insert the First OS Disk (pa) and then press <enter>".

Put the OS CDROM Disk 1 of 2 into the CDROM Drive. Wait for 20 seconds, or until the amber light on the CD Drive stops blinking, then push the **<Enter>** key.

7. After an additional 3-6 minutes, the system will again prompt with:

"Please Insert the Second OS Disk (pb), then press <enter>"

Put the OS CDROM Disk 2 of 2 into the CDROM Drive. Wait for 20 seconds, or until the amber light on the CD Drive stops blinking, then push the **<Enter>** key.

8. Loading the rest of the Operating System off the second disk will take about 30 minutes with and will not require any user intervention.

.

.

When finished, the system will stop at the "#" prompt. Type the following to continue.

.

exit <Enter> (to exit back to the original inst> prompt.)

.

.

Inst> **exit** <Enter>

.

.

Automatically reconfiguring the operating system.

.

.

.....Ready to restart.... **y** <Enter> (Restarts the system)

.

.

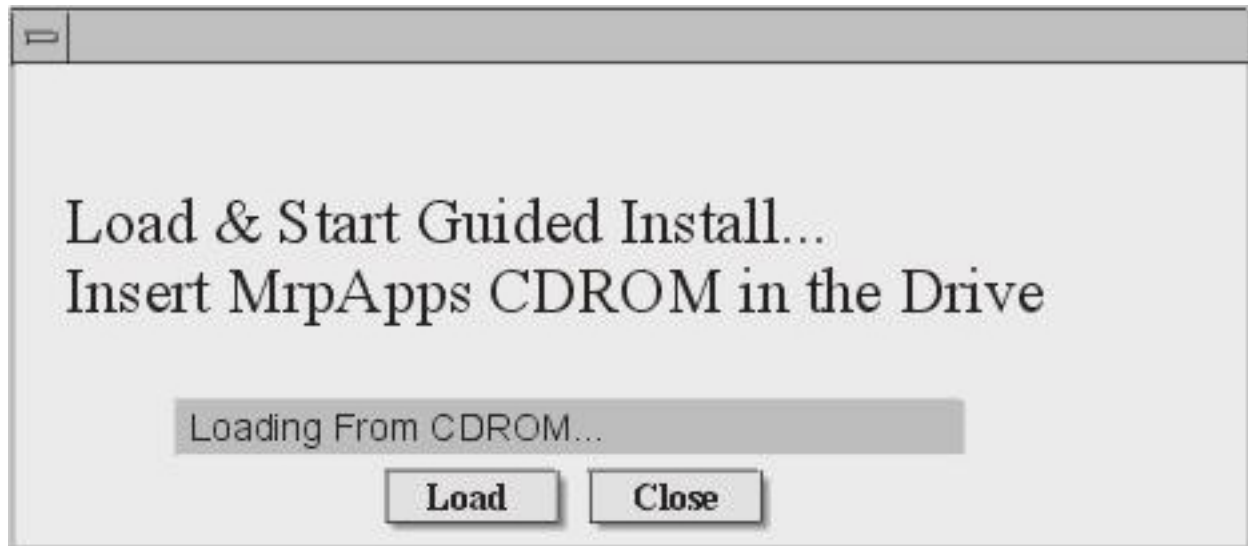
This will cause the system to finish some processing and reboot. The OS load portion of the Software Install is complete.

Important!
Do not log into EZ Setup. It is no longer used as part of Install.

9. The OS load is complete. Continue to Section 6 of this document for the MR Apps load.

6- HOST MR_APPS SOFTWARE LOAD

1. When the system is finished rebooting, **log in as root**. There is no password at this time. You will first see a startup box that will asks for the MrpApps CDROM. See Illustration 6-1.



GUIDED INSTALL LOAD SCREEN
ILLUSTRATION 6-1

2. Remove any CD from the CDROM drive and insert the MrpApps CDROM into the CDROM drive. Wait 20 seconds or until the amber light on the CDROM drive stops blinking and push the **<Load>** button. The Guided Install will load and start.

Note

Help documentation is built into the Guided Install GUI. Push the **<Help>** button on the tab you need information for in the GUI. When the GUI first starts press the **<What's New>** button for information on using the GUI.

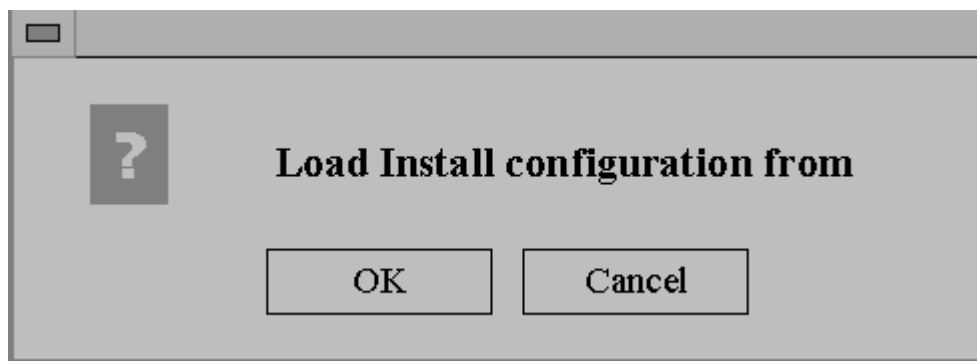
3. On initial startup, a popup box will appear over the Guided Install interface describing the tool and its use. After reading the information push the **<OK>** button to continue.

4. Next, a popup box (see Illustration 6-2) will appear over the Guided Install interface. If you are reloading 8.3 or greater software, you can insert the SaveInfo MOD (created at the start of this procedure) and push the **<OK>** button and your last configurations will be reinstalled. You can verify by checking each tab.

Important: The last Install GUI configuration can only be installed if it was saved previously by putting a SaveInfo MOD into the drive and from the Install GUI menu bar, **<File>** then select **<Save To MOD>**

5. If this is an upgrade from 8.2.x software, push the **<Cancel>** button to continue and continue manually with the install.

Important: Do not logoff or reboot the system before exiting the Install GUI. If you log out before exiting the Install GUI a 34 Mbyte core file will be generated in “/usr/g/service/log”. If this should happen, you will need to go to “/usr/g/service/log” and remove the core file.



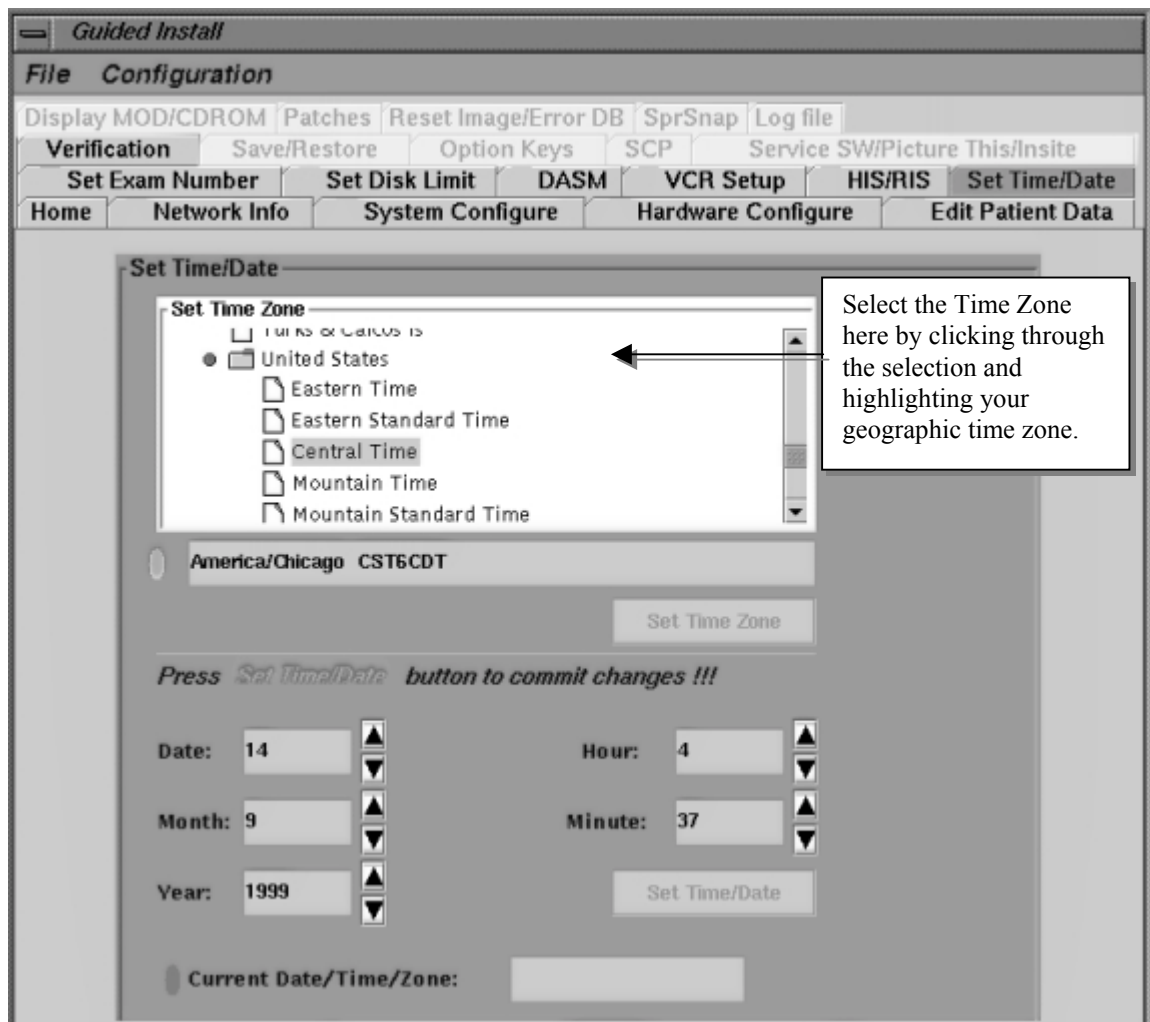
INSTALL CONFIGURATION POPUP SCREEN
ILLUSTRATION 6-2

6. The Guided Install home screen is displayed. See Illustration 6-3. Do the following to continue with the MRApps load:

Press the **<Next Tab>** button to move through active tabs from Network to Set Time Zone. Fill out all empty entries or make any updates as appropriate. This information should have been saved in Table 4-1 in section 4-2 of this document. When you reach the “Set Time / Date” tab, go to the “Set Time Zone” box and use the selections to set the time zone your system is in. See Illustration 6-4.



GUIDED INSTALL - HOME SCREEN
ILLUSTRATION 6-3



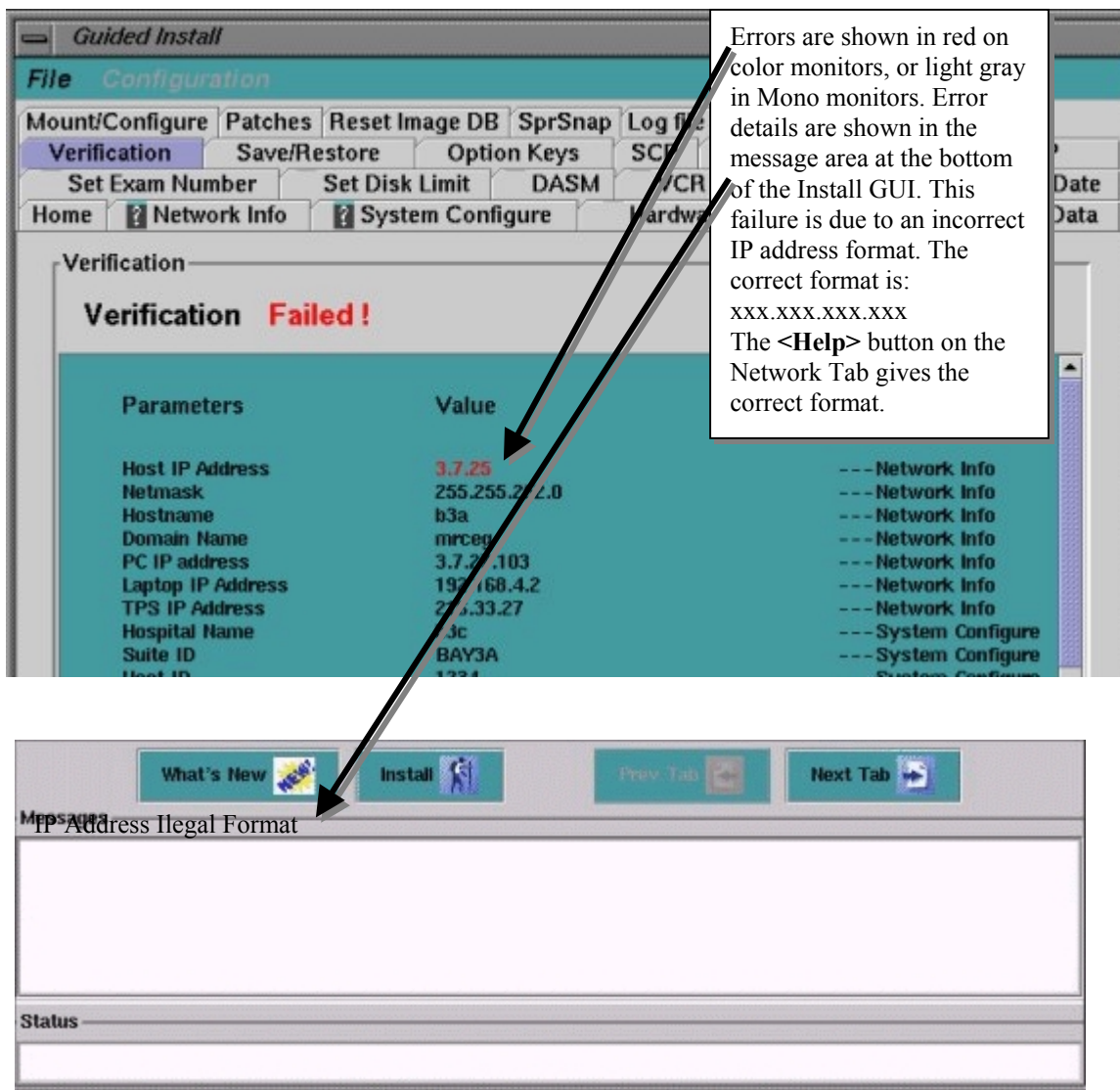
GUIDED INSTALL - SET TIME/DATE SCREEN
ILLUSTRATION 6-4

7. Proceed to the verification tab. Once in the Verification tab, look at pass or fail status of entries. If any entry has a failed status, the **<Install Software>** button will not be active. Failed entries will be due to missing entries, or entries not in the correct format. Go to the tab with the error and correct it. Pushing the **<Install >** button will again open the Verification tab. If all entries pass verification at this point, push the **<Install Software>** button again to begin the MR Apps load. This will take 20-30 minutes, depending on the machine being loaded.

Note

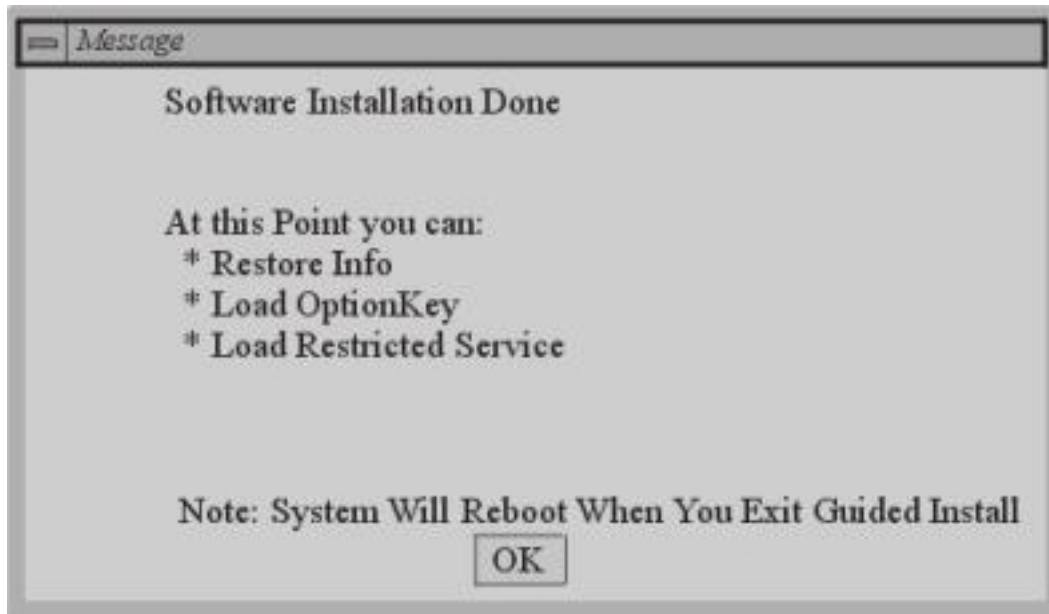
If YP is turned on, the system will take longer to boot (2 to 4 minutes). Turn it on only if the site external network the system is being installed on needs YP on. Be aware there is a small delay that may not be seen on other sites.

It is not necessary that any one tab be filled out in any particular order. Just all required configuration tabs must be filled in or the install will not continue. The Verification Tab will list any formatting errors as well as current settings. See Illustration 6-5 for an example of a failure shown in the Verification Tab in networking. The Host IP Address has an incorrect format.



GUIDED INSTALL - VERIFICATION SCREEN
ILLUSTRATION 6-5

8. Once the MR Apps load is done, the popup in Illustration 6-6 is displayed.



GUIDED INSTALL – SW DONE SCREEN
ILLUSTRATION 6-6

9. Push the **<OK>** button to close the popup. At this point the GUI will change from the install phase to the Configure phase. If a SaveInfo info MOD is available for this site, proceed to the Save/Restore tab and Restore the system info. If no SaveInfo MOD is available, go to each tab in the Install GUI and fill in or update all system information manually. A help button is available on each tab, which will open a popup box with information on the required data and its format for that page.
10. When the RestoreInfo is finished a popup will open telling the installer that upgrade scripts must be run. See Illustration 6-7.
11. Push the **<OK>** to continue. The upgrade script menu will open (see illustration 6-8).



GUIDED INSTALL – UPGRADE SCRIPTS MESSAGE
ILLUSTRATION 6-7

This script upgrades the system config files
For LFC software. After Restore Info is completed,
the following UP scripts must be run before scanning.

- | | |
|--------------------|--|
| 1. ConfigUP | Update MRconfig.cfg/GradientConfig.cfg files |
| 2. CoilUp | Update CoilConfig.cfg file |
| 3. Protocol1Up | Upgrades 8.2.5/83 Protocols |
| 4. ProtFormat | Reformats site protocols names |
| 5. Restore_License | Must be run individually |
| 6. Run all scripts | Except Restore_License |
| 7. Exit | Return to Guided Install. |

GUIDED INSTALL – UPGRADE SCRIPTS MENU

ILLUSTRATION 6-8

You must always run these updates in the following order:

- Type: **6 <Enter>** This will take about 1 minute to run.
- Type: **5 <Enter>** This will take < 1 minute to run.
- Type: **7 <Enter>** This will close the menu and return to the GUI.

Note

Refer to Appendix A for parameters updated by the Configup scripts

- Once back on the Guided Install screen, use the cursor to select **< File>** and then **<Quit>** at the top left of the GUI. When the GUI shuts down, the system will reboot and all changes restored or made will be set.
- After the system has finished rebooting, once again log in as **root**. Password is **operator**.
- You will be asked once entering the root desktop if you want to start the Install GUI. Push the **<Yes>** button.
- When the install GUI opens, go to the Load ServiceSW/Picture this/InSite tab and install service software. GEMS Field Engineers loading GEMS Restricted Service Software will also configure InSite Interactive at this time (see Illustration 6-9). Follow the instructions given in popup windows as presented.

InSite Interactive Platform Configuration

Information ProDiags Health Page **Device Connection** InSite Checkout

Connection Device Configuration
Last Update: Fri Aug 13 15:50:42 CDT 1999

Device Connection Type: Modem

Select the InSite Connection Type above, or select the Modem settings given below and Press
APPLY – To update the modem and modem configuration files with model and country defaults.
CUSTOM – For other modem settings.

Dial-out Prefix: 8, Dialing mode: Tone

Modem Type: USRobotics Courier V.34 or V.Everything Country: Default – All Others

CPU Serial Port Name: /dev/ttyf1 CPU Serial Port Speed: 115200

APPLY CUSTOM >>

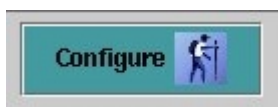
Status: IIP Configuration Tool Ready for Input... EXIT

CONFIGURE INSITE MENU
ILLUSTRATION 6-9

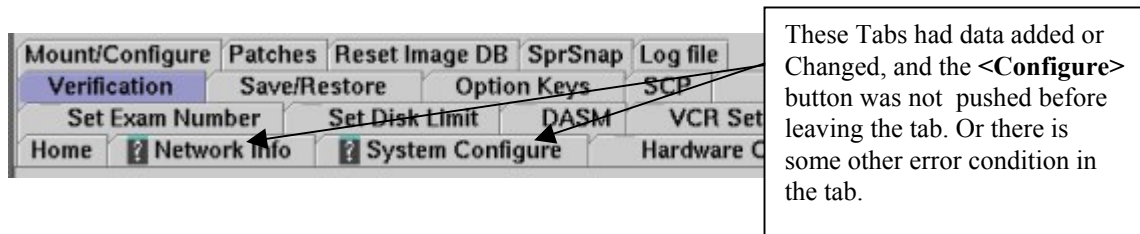
- After service installation, move through and check accuracy of each tab. Insure all system options for the site are active. If needed, load software option keys at this time. If the site has any of the following option keys, they must always be reloaded, as they are not saved with SaveInfo: *IVI, Voxtool, FuncTool, Clairview*.

Note: Version M4 software comes with two software MOD keys that will always need to be loaded. These two MOD's (two sides on one MOD and one side on the other MOD must be loaded) contain the LX Scantools software that is shipped with every system.

- After information as been restored from MOD or changed in a tab, and configuration updates have been run, push the **<Configure>** button on the bottom of the tab to apply the configuration.



If the **<Configure>** button is not pushed to commit changes to a tab, or an error is present in a tab, a question mark will be seen on the tab, as shown by the following.



18. Proceed to load or configure any options that may not have been restored by the SaveInfo disk. Go to the Load Service tab and load any service software. Update any patches required. Perform a SaveInfo to save the latest changes.
19. Exit the install GUI by selecting **<File>** then **<Quit>** from the top of the interface. If any error conditions still exist, you will be warned that no changes will be made. If no error conditions are found, the GUI will save user parameters.
20. Reconnect the Network interface at this time if disconnected at the beginning of the install. Also turn the DASM back on if disabled at the beginning of the install. Log out of root and reboot the system.
21. This concludes the software load. Continue with section 8 to update the monitor gamma settings.

7- MONITOR GAMMA VALUE RESET

After a complete software load has finished, do the following steps to update the monitor configurations. This insures compatibility between monitor and camera.

1. In the C-Shell, type **su <Enter>** to log in as root; enter the root password (default is **operator**).
2. **B/W Monitor:**
 - a. Type **jot /usr/g/grayPix/grayPix.cfg <Enter>** to edit the grayPix.cfg file.
 - b. Modify the numerical value for “gamma” to **0.9**

Color CRT Monitor:

 - a. Type **jot /usr/g/colorPix/colorPix.cfg <Enter>** to edit the colorPix.cfg file.
 - b. Modify the numerical value for “gamma” to **1.1**

Color LCD Monitor:

 - a. Type **jot /usr/g/colorPix/colorPix.cfg <Enter>** to edit the colorPix.cfg file.
 - b. Modify the numerical value for “gamma” to **1.05**
3. In the jot window, select **File -> Save**, then **File -> Exit** to save the changes. This modifies the gamma value such that if the system reboots, the value is not lost.

8- FINAL CONSIDERATIONS

8-1 Required Calibrations

The following calibrations are required after loading Release 8.3 M4 software.

- ☐ **Run SPT Full Test Mode** (Head and body but not LVShim) to check overall system performance.
Update Gradcal if out of spec.
- ☐ **Perform another “Save INFO”** to save the latest changes (refer to Software Install SGI Host procedure, Section 2-2-1).

8-2 Bore Temperature Sensor - Must Be Working!!

If the bore temperature sensor is disabled, sites with iDrive software will have problems. Sites running the iDrive software must have a functional bore temperature sense circuit. The symptom for a non-functional Bore sensor will be evident when an iDrive scan is attempted; the prescan will complete OK, but the scan will stop immediately after the first pulse is heard in the body coil. No images will be produced and no errors will appear in the log.

If the system has iDrive software (as listed in *Option Key* tab of Guided Install), to verify functional bore temperature circuit, set up the following scan (or any other Real Time scan). Then download the scan and select **[Scan]**. If prescan completes and the scan continues to run, you have a functional bore temperature circuit. If prescan completes OK, but the scan stops immediately after the first pulse, check the bore temperature circuit.

The Release 8.3_M4 upgrade is complete!

iDrive sample protocol

<u>PATIENT REGISTER</u>	[New Pt]
<u>PATIENT INFORMATION</u>	
Patient Id	geservice
Patient Name	test
Weight (Lb)	111
	[Landmark]
Landmark	[Sternal Notch]
<u>PATIENT PROTOCOLS</u>	[Patient Position]
<u>PATIENT POSITION</u>	
Patient Position	[>] [Supine]
Patient Entry.	[>] [Head First]
Coil	[...] [Body] [Accept]

IMAGING PARAMETERS

Plane	[>] [Axial]
Mode	[>] [2D]
Pulse Seq	[...] [FGRE] [Accept]
Imaging Options	[Real-time]
Psd Name	no entry
Protocol	no entry

SCAN TIMING

# of Echoes	1 (default)
TE	[Mn Full]
TR.	[Mn]
Flip Angle	[20]
Bandwidth	[62.50]

SCANNING RANGE

FOV.	[32]
Slice Thickness.	[7]
Spacing	[1.5]
Start	0
End..	0
# Slices	1 (default)
L/R Center	0 (default)
P/A Center	0 (default)
Table Delta	0.00 (default)

ACQUISITION TIMING

Freq	[128]
Phase	[128]
NEX	[1]
Phase FOV	1.00 (default)
Freq Dir	[>] [R/L]
Auto Center Freq	[>] [water]
Autoshim	[on]
(lowest window)	[Save Series]

APPENDIX A - “CONFIGUP” SCRIPT MODIFICATIONS TO FILES

When RestoreINFO is run after loading new Release 8.3 M4 software, new version configuration files are overwritten. The “configup” script updates the MR System configuration file with the new 8.3 M4 changes as needed:

1. The “configup” script updates the MRconfig.cfg file with the new “fastprescan” parameter if it is not present (if “fastprescan” parameter is present, it does not add or change it):

```
# This field is added by shivar@wge for MRIge47785
# A fastprescan field to turn Faster Prescan on & off .
#           type = integer, legal values = 0 (off)
#                                           = 1 (on)
fastprescan = "1"
```

2. The “configup” script updates the MRconfig.cfg file with the new “cerdType” parameter if it is not present (if “cerdType” parameter is present, it does not add or change it):

```
# Type of (Universal) Combined Exciter Receiver
# Data Acquisition Board (CERD)
# type = integer
# Usage
# X Y Y
# \ \ \----- (U) CERD Slot #1 (VME slot 13 , right next to
# power supply)
# \ \----- (U) CERD Slot #2 (VME slot 10)
# \----- MKE or YMS Receiver mapping
# Where X =
#   0: MKE receiver mapping scheme
#   1: YMS receiver mapping scheme
#
# Where Y =
#
#   0 : No(U) CERD Installed
#   1 : Narrowband CERD
#   2 : Universal CERD (UCERD)
# Legal Combinations
#
#   0yy: MKE default receiver mapping
#   1yy: YMS default receiver mapping
#   x01: Narrowband CERD on Slot #1
#   x02: Universal CERD on Slot #1
#   x21: Narrowband CERD on Slot #1 ;Universal CERD on Slot #2
#   x22: Universal CERD in both Slot #1 and Slot #2
cerdType = "001"
```

3. The “configup” script updates the GradientConfig.cfg file with the new Full Power SGD parameters if not present (if the following parameters are present, it does not add or change them):

```
# Added by shivar@wge for MRIge49945
```

```
Sdlim = "0.60"
```

```
Lcoil = "1.40"
```

```
Rac = "152"
```

```
Pmgs = "3000"
```

```
mGRAMpower = "9000"
```

APPENDIX B - HOST SOFTWARE INSTALL WORKSHEETS

Host IP Address: The Host IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

IWS PC IP Address: The IWS PC IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

FE Laptop IP Address: The Laptop IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

TPS Subnet Address: The TPS is on a subnet of the Signa LX scanner. A default address is loaded during the load from cold process and it should not be changed.

Hostname: This is the network name given to the scanner host computer.

Hospital Name: The Name of the hospital or other facility where the Horizon LX is installed. Name can include characters and/or integers up to 32 characters.

Doctors Title: The title to be used at the scanning site by the reading physician (Radiologist or Diagnostician).

Suite ID: Name of the Suite (if any) the scanner will be connected to. Leave blank otherwise.

Unique ID: The Unique System ID is a 16 alphanumeric character string assigned at the time of the system is first installed. The Unique system ID is assigned by GE CARES when the first dispatch is opened to document the start of installation.

Service ID: The Service ID is a 16 alphanumeric character string assigned at the time the system is first installed. This is the same number that GE CARES uses to track financial and administrative data for the system.

Weight Unit: The weight system to be used by the scanner. Kilo Grams or US pounds.

Monitor type: The Type Monitor the scanner will be using.

Windows 95 OEM Number: The Microsoft OEM Number found on the "Proof of Authenticity" for Windows 95. _____

NetMask: For new installs, the Netmask should be obtained from the site network administrator if the scanner will be connected to the sites internal network. If this is a stand alone system, make no changes to the default netmask installed during load from cold. For existing systems, the netmask is not on the Install GUI (requires System Shutdown). Get this information after all other information has been gathered as follows: **[System Shutdown]**, Log in as Root, type **operator <Enter>**, **[Reconfig]**, **[Basic Network Setup]**.
Record Netmask here: _____

Check (✓) the Options installed at this Site on the System Options Table below.

TABLE B-1
SYSTEM OPTIONS DATA TABLE

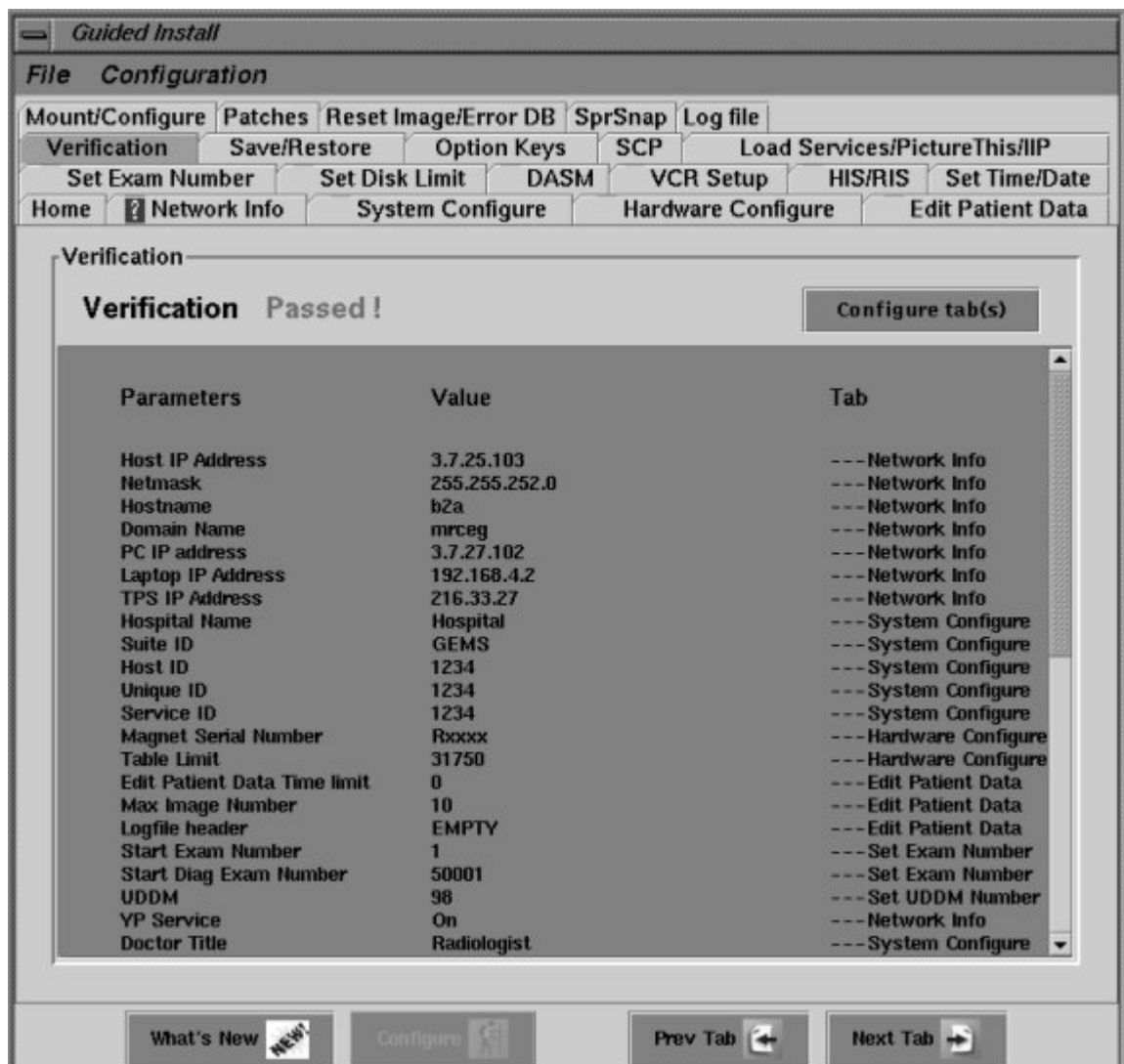
	Options For This Site		Options For This Site
	Fast Spin Echo & FLAIR		Func Tool
	Fast Gradient Echo		HIS RIS
	Cine		E3D TOF
	Phase Contrast Vascular Imaging		FSE-XL
	Videoconferencing		Clariview
	Multi-Slice/Multi-Angle Oblique		Bloodsupp
	Echo Planar Imaging		Fastcine
	Time of Flight		Tagging
	Spectroscopy/PROBE		Sgdperf
	Research		DW-EPIEZ
	Research PSD		
	Proactive Service		
	3D Surface/Voxtool		
	Interactive Vascular Imaging		
	SGD - EchoSpeed		
	DW EPI		
	FLAIR EPI		
	SPECIAL		
	Smart Prep		
	SSFSE		
	Three Plane Localizer		

APPENDIX C – RECONFIGURING THE SYSTEM USING THE INSTALL GUI

If the system needs to be re-configured at some later time, do the following to start the Configuration GUI:

With Signa up and running, go to the service desktop and push the **<Install>** button on the toolbar. A C-Shell will open and ask for root password. “**operator**” **<Enter>** is the default. The Install GUI will load and start.

Help documentation is built into the GUI. Push the **<Help>** button on the tab you need information for in the GUI. Push the **<Next>** to move through tabs in order. Modify entries to match your system as required. When satisfied, push the **<Configure>** button at the bottom of the tab page. Or wait until all tabs have been modified and go the Verification tab. Review changes that are to be made are correct, then push the **<Configure Tab(s)>** to update the system.



Before exiting the GUI, insert the your SaveInfo MOD or new MOD into the drive. Go to the top left corner of the Install GUI and select **<File>** and then **<Save to MOD>**. This process does not create a SaveInfo disk. It just creates a copy of the information already entered in the GUI tabs for later use for updating the tabs in a new install.

Exit the install GUI by selecting **<File>** then **<Quit>**. If any error conditions still exist, you will be warned that no changes will be made before exiting.

Note

The GUI will not automatically reboot the system in configure mode. It will be up to the operator to reboot after making configurations.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Nov 3, 1999	J. Wolak	Initial version for Release 8.3
0	Nov 17, 1999	J. Wolak	Product release version



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